ZOO403

MID TERM SUBJECTIVE BEST OVER ALL FILES

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Insight learning:

A type of learning that uses reason, especially to form conclusions, inferences, or judgments, to solve a problem. It is based on advanced perceptual abilities such as thought and reasoning.

EXAMPLE: Kohler found that chimpanzees could use insight learning instead of trial-and error to solve problems. In one example, a banana was placed high out of reach that the chimpanzees found a way to reach it. They stack boxes on top of each other to reach it and used sticks to knock the banana down.

what are the types of the learning behaviour in biology.

Habituation, imprinting, classical conditioning, operant conditioning, and cognitive learning.

what are the feature detection:

Feature detection is a process by which the nervous system sorts or filters complex natural stimuli in order to extract behaviorally relevant cues.

The three major groups of so-called feature detectors in visual cortex include simple cells, complex cells, and hypercomplex cells.

Animal Orientation: "Ability of an animal to determine its position in space and move predictably with respect to specific stimuli"

Animal Orientation: Examples:

1=Standing up after falling down (orientation against gravity),2=Orienting towards incoming light, 3=Acoustic orientation, 4=Olfactory orientation

Sensory Motor Basis of Orientation:

1=Animal orientation is a complex process, 2=Requires integration of sensory input and coordination of motor output.

Input-Output Integration:

Integration includes:1= Receiving information through sensory receptor neurons. 2=Processing and correlating it in the CNS. 3=Forming a response through muscles that cause orienting movement.

Difference between morphological and behavioral responses.

1. Morphological Responses

- changing in the of the body appearance
- Growing thicker fur or change in fur color in winter

Examples:

Chameleon change color when it feel threatened.

2. Behavioral Responses

- changing behavior to adapt to the change
- moving to a more favorable location
- Cooperative behavior
- Agonistic behavior when threatened

Examples:

During the winter season, Houbara bustard finds refuge in plain lands of Sindh, Southern Punjab and Baluchistan.

Habitat of dolphin:

Dolphins are found all over the world; generally in shallow cold waters as well as the warm tropical waters.

Types of animal behavior:

There are 2 basic types of behavior.

1= Innate behaviors or Inherent behavior 2=Learned behaviors or acquired behavior.

• Nature/innate [Instinct or genes determine behavior]

• Nurture/learned [Experience and learning influence behavior]

Names of Organisms in which Researchers examined Cognition:

Researchers have examined animal cognition in:

- mammals (primates, cetaceans, elephants, dogs, cats, horses, cattle, raccoons and rodents),
- birds (including parrots, fowl and pigeons),
- (lizards, snakes, and turtles)
- Fishes
- Reptiles
- Invertebrates (cephalopods)

Explain instinctive behavior with the example of birds how they make their nests:

Innate behavior is also called instinctive behavior. **Instinct** is the ability of an animal to perform a particular behavior in response to a given stimulus the first time the animal is exposed to the stimulus. In other words, an instinctive behavior does not have to be learned or practiced.

A bird nest is the spot in which a bird lays and incubates its eggs and raises its young. For most birds, when a male and female form a breeding pair nest building activities will follow.

Nest Building Behaviour: A nest site is first selected and nest building begins. This can involve both birds as a pair or either sex alone. Different species of birds build different nests. Some nests are very simple constructions, whilst others are more complex. Birds usually only build nests for holding their eggs and rearing chicks.

What is associative behavior:

Associative Learning “A learning that involves making associations between experiences”

Example: A mouse bites a brightly colored caterpillar, It tastes bad

Waggle dance:

A special dance performed by a worker bee called “waggle dance”

Neuron components

Neuron has three components: 1= Cell body 2= Axon (moving nerve impulse away from cell body) 3= Dendron (moving nerve impulse toward the cell body).

Imprinting:

“Formation of a longlasting behavioral attachment to a particular individual or object at a specific stage in life”

• A type of learning behavior • A very young animal develops attachment with the very first object it has visual, auditory or tactile experience • It thereafter follows that object

Examples:

• Seagulls • Sensitive period for a parent to bond with its young lasts one to two days • If bonding does not occur, parent will not care for the infant, leading to death of the offspring.

nervous system role in behavior:

nervous system controls complicated processes like movement, thought and memory. It also plays an essential role in the things your body does without thinking, such as breathing, blushing and blinking.

reflex arc components:

A sensory organ (receptor), A sensory nerve (afferent nerve), The spinal cord or brain, An intermediate motor or efferent nerve, A motor organ or effector.

Reflex action: Reflex action is a sudden and involuntary response to stimuli.

Anolis lizard hormone:

Anolis Lizards Testosterone is critical for stimulating male sexual behaviors. Castration reduces the display of male reproductive behaviors, and testosterone can prevent or reverse that effect.

Chemical signals advantages: Effective over distances and around corners. Effective either day or night. Longer lasting than visual or auditory signals. Metabolically “inexpensive” because only small quantities are needed.

What is taxis: “An oriented movement toward (positive taxis) or away (negative taxis) from some stimulus”. Taxis—Examples: • River fishes (e.g. trout) swim or orient in an upstream direction (toward the current).

Types:

1. Positive taxis If the orientation is towards the stimulus it is called as positive taxis.

2. Negative taxis If the orientation is away from the stimulus it is known as negative taxis

Kinesis: “a change in activity or turning rate in response to a change in location”.

Significance of Habituation: • An animal conserves energy and time by habituation to unimportant stimuli • Energy and time are better spent on other important functions.

What do you know FAPs in cocoon spinning behavior:

During spinning, turning the entire body around. The spinning behavior includes two distinct and mutually exclusive movement patterns (FAPs) i. the stretch-bend ii. the swing-swing.

characteristics of fixed action patterns. FAPs are:

- Stereotyped
- Complex
- Species-characteristic
- Released
- Triggered
- Independent of experience

Objectives of Ethology:

The aim of ethology is to explain both phylogenetically and physiologically the functional relationships of all factors involved in behavior.

Classical conditioning: An arbitrary stimulus becomes associated with a particular outcome. Example Experiments of Russian physiologist Ivan Pavlov.

Operant Conditioning: "An animal learns to associate one of its own behaviors with a reward or punishment and then tends to repeat or avoid that behavior" also called trial-anderror learning.

Example: An animal may repeat an action that has resulted in a surprise finding of food.

Characteristics of innate behavior:

• Pattern of behavior is inherited. It is passed on from parents to offspring • It is unlearned behavior. • It occurs in all the members of a species hence it is species specific and predictable. • It is not dependent on past experience as it is an inborn response to a stimulus. • It takes place in individuals even when kept in isolation away from their fellow members. • Innate behavior has high adaptability and survival value.

Synapse:

synapse, also called neuronal junction, the site of transmission of electric nerve impulses between two nerve cells (neurons) or between a neuron and a gland or muscle cell (effector). A synaptic connection between a neuron and a muscle cell is called a neuromuscular junction.

process of panting:

panting, a method of cooling, used by many mammals, most birds, and some reptiles, accomplished by means of the evaporation of water from internal body surfaces.

what is the phermomes: • Chemical messengers released by many animals in environment to communicate • Spread through air • Affect the behavior of individuals of a species.

physiological response:

Physiological responses are the body's automatic reactions to a stimulus. Most of us are familiar with the automatic and instinctive physiological responses we experience every day, but we typically remain unaware of them.

web making process in spiders:

Spider silk is made of connected protein chains that help make it strong, along with unconnected areas that give it flexibility. It is produced in internal glands, moving from a soluble form to a hardened form and then spun into fiber by the spinnerets on the spider's abdomen.